Paul Schulte

p-schulte.github.io | paul.schulte@rwth-aachen.de | LinkedIn

Education

RWTH Aachen University — Oct 2025 – Present
M.Sc. Computer Science — Aachen, Germany

- Specialization planned in Machine Learning, Optimization, Algorithms, and Graph Theory

RWTH Aachen University — Oct 2022 – Aug 2025
B.Sc. Computer Science — Aachen, Germany

- **GPA**: 1.1 (with distinction; 1.0 best)
- **Dean's List**: Top 5% in academic years 2022/23 and 2024/25
- **Accelerated degree**: Completed in 5 out of 6 semesters
- **Key Coursework**: Machine Learning and Data Science (1.0), Data Structures and Algorithms (1.0), Computability and Complexity (1.0)
- **Bachelor Thesis**: *Action Classification Through Dynamic Scene Graph Generation from Image Sequences* (Grade: 1.0)

Aalto University — Aug 2024 – Dec 2024
Semester Abroad (Computer Science) — Espoo, Finland

- **GPA**: 5.0 / 5.0
- **Coursework**: Graph Theory, Machine Learning, Competitive Programming, Distributed Algorithms

Research Experience

RWTH Aachen University, Chair for Machine Learning and Inference (i6) — Aachen, Germany
Student Research Assistant — Sep 2025 – Present

- Extended Bachelor thesis research on dynamic scene graphs for action classification
- Equal-contribution first author on a conference paper submitted to ICAPS 2026

Robotics Institute, Carnegie Mellon University — Pittsburgh, PA, USA
Undergraduate Research Intern (RISE Worldwide) — May 2024 – Sep 2024

- Conducted research in control theory, trajectory planning, and autonomous driving
- Developed a Model Predictive Control (MPC) framework with Control Barrier Functions (CBFs) for safety-critical planning
- Implemented real-time MPC in JAX on CPU-based F1TENTH hardware, reducing reliance on GPU-heavy approaches

Professional Experience

Siemens Logistics — Düsseldorf, Germany
Software Engineering Working Student — Feb 2023 – Nov 2024

- Designed and implemented C# features for a large-scale airport baggage handling system planning tool
- Optimized computationally intensive components, achieving runtime reductions of up to 80%

Selected Projects

Math Exercise Generator | *JavaScript, HTML, Vue.js* — 2023

- Built a free web-based linear algebra exercise generator with automatic solution computation

Neuroevolution of Fixed Topologies | *Python, Reinforcement Learning* — 2021

- Implemented neural networks and evolutionary algorithms from scratch
- Applied methods to OpenAI Gym environments and documented results in a technical report

Hanging Wall Plotter | *Python, Raspberry Pi* — 2019

- Designed and programmed a two-motor wall-mounted plotter
- Converted PNG images to G-code and executed motion control

Honors & Distinctions

DAAD RISE Worldwide Scholarship 2024
- Competitive research scholarship for international STEM internships

German Mathematical Society (DMV) 2022
- Student membership awarded for outstanding performance in mathematics during high school

German Physical Society (DPG) 2022
- Student membership awarded for outstanding performance in mathematics during high school

Technical Skills

Languages: German (native), English (C1)
Programming: Python, C++, Java, C#
Tools: Linux, Git, Docker, LaTeX